

PANSHUL JINDAL

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AI engineer who ships agent systems to production. At Microsoft I built the agent runtime other teams build on, and the queue that lets an agent hand off to its future self so long-running work survives restarts. I care as much about the quickstart as the architecture.

EXPERIENCE

Software Engineer - Microsoft Corporation (Copilot Growth)

Oct 2025 – Present

- Recognized as a **Top 30 AI power user across Microsoft** - company-wide.
- **Lead AI architecture** for Iris, one of the world's largest marketing platforms - set the runtime, orchestration patterns, and evaluation bar now adopted by 5 teams across 14 services.
- Built an **autonomous campaign-launcher** that assumes the role of a human campaign owner, coordinating specialized agents (planner, content, validation) over typed contracts to translate raw intent into live deployments with 100% less manual configuration.
- Developed a fault-tolerant **agent state machine and queueing system**. By serializing execution steps, agents can safely yield and resume in fresh instances, ensuring long-running, multi-day campaigns survive infrastructure restarts seamlessly.
- Led the rollout of **code memory** across our services, recording each agent's reasoning for a change alongside the change itself - so context survives the session that produced it, and the next agent or engineer inherits why the code looks that way instead of re-deriving it.
- Own the delivery platform the agents send through - 5 microservices handling **~1B** emails and push notifications weekly. Added idempotent send-path retries, backpressure, and probe-based automated pod recovery.

Software Engineer - Visiostack

Jun 2025 – Sept 2025

- Built a **retrieval-augmented generation (RAG)** pipeline leveraging OpenAI embeddings and ChromaDB vector store to convert natural language queries into structured JSON outputs aligned with a given schema.
- **Fine-tuned** a domain-specific model via supervised fine-tuning on synthetic data (generated using **OpenAI's batch API**), boosting query accuracy and response consistency for the enterprise search application.

Research Assistant - RailTEC, UIUC

Jun 2024 – May 2025

- Researchers had terabytes of sensor and image data and no way to ask it questions. Built the **natural-language-to-SQL** interface so they could query it directly instead of filing requests with an engineer.
- Paired PySpark anomaly pipelines with **LLM analysis** of fault reports so detections arrived with an explanation rather than a score, and shipped it as modular Azure microservices instead of leaving it in notebooks.

Software Engineer - Kwipic AI Solutions

Jan 2023 – Dec 2023

- Owned notification personalization end to end - LLM-driven per-user content across push and email, in early 2023, when that meant raw completions with no frameworks and no tooling.
- Designed the auth and photo-delivery APIs consumed by the iOS, Android, and web clients: versioned endpoints with signed-URL delivery, serving over **10 million users** across platforms.
- Took the **facial-recognition engine to 98.9% accuracy** and cut image-preprocessing latency by over 30%, in the hot path of every photo delivery.

SKILLS

Languages – C#, Python, Go (Golang), Java, SQL, TypeScript

AI and Agents – Agent runtime and Loop Design, Microsoft Agentic Framework (MAF), Hugging Face Transformers, LangChain, LangGraph, LLM Fine-tuning (SFT, PEFT)

Product & Web – FastAPI, Node.js, REST and gRPC API design, SDK and library design, versioning and backward compatibility, developer docs and quickstarts

Infrastructure & Cloud – Kubernetes, Docker, GitHub Actions, CI/CD, Redis, PostgreSQL, Datadog, Azure, AWS

EDUCATION

University of Illinois, Urbana Champaign

Jan 2024 - May 2025

Masters in Computer Science

CGPA: 4.0/4.0

Relevant Coursework: Machine Learning for Software Engineering, Artificial Intelligence, User-Centered ML

PROJECTS

TaskPilot: Agentic Workflow Engine for Autonomous LLM-Orchestrated Tasks (FASTAPI, Docker, OpenAI API, Redis)

- Architected a modular multi-agent system where autonomous AI agents plan, execute, and validate tasks using tool integrations.
- Built Redis-backed task queues with FastAPI and a real-time memory store, enabling scalable automation with persistent context.

AutoTest: AI-Powered Code Coverage Booster (DeepSeekCoder, CI/CD, Pytest)

- Engineered a tool leveraging DeepSeekCoder to analyze code and synthesize unit tests, raising coverage from **~40% to 100%**.
- Built an **auto-refinement loop** where failing tests trigger iterative adjustments (using execution logs) to improve subsequent outputs.

LEADERSHIP & COMMUNITY

Technical Lead - Apple Developers Group, VIT

- Led AI-focused workshops and mentorship programs for **200+ students**, providing hands-on training in machine learning applications.
- Spearheaded the creation of internal tooling and documentation standards to streamline project collaboration for new contributors.